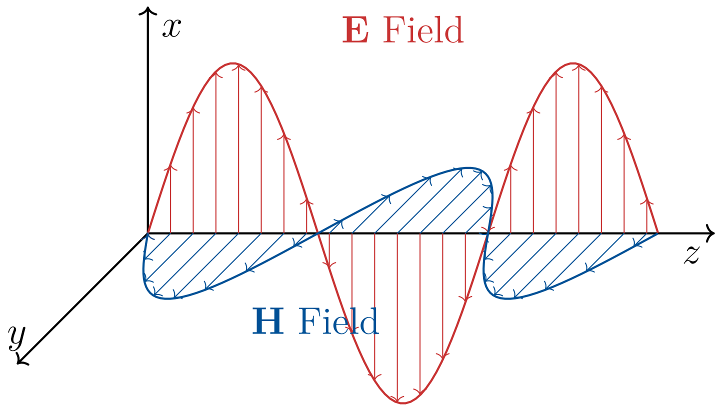
Comprehensive Solutions Manual

Electromagnetic Field Theory

B.E. Electrical Engineering
Second Year, Second Semester - 2024 Exam



Detailed Step-by-Step Mathematical Formulations and Vector Calculus Proofs

Prepared as an Exhaustive Study Resource

Contents

List of Figures

PART I: Electrostatics & Steady Fields

1 Question 1 - Charge Distributions and Potentials

1.1 (a) Uniform Electric Field over a Unit Cylinder (5 Marks)

Problem Statement:
A cylinder of unit volume is placed in a uniform field with its axis parallel to the direction of the electric field. Determine the total charge enclosed by the unit cylinder.

Detailed Step-by-Step Solution:

1. **Theoretical Foundation:**
To determine the total charge enclosed by any closed surface, we rely on the fundamental postulate of electrostatics, known as **Gauss's Law**.

2. **Mathematical Formulation of Gauss's Law:**
Gauss's law in its integral form relates the electric flux leaving a closed surface S to the charge enclosed Q_{enc}:

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enc}} \tag{1}$$

where $\mathbf{D}$ is the electric flux density vector, and $d\mathbf{s}$ is the differential area vector (pointing outward).

By applying the **Divergence Theorem**, we can convert this surface integral into a volume integral over the volume v enclosed by S:

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_v (\boldsymbol{\nabla} \cdot D)\, dv \tag{2}$$

Equating the two expressions gives the differential form of Gauss's Law:

$$\boldsymbol{\nabla} \cdot D = \rho_v \tag{3}$$

where ρ_v is the volume charge density. Therefore, the total enclosed charge can be expressed as:

$$Q_{\text{enc}} = \int_v \rho_v\, dv = \int_v (\boldsymbol{\nabla} \cdot D)\, dv \tag{4}$$

3. **Applying Medium Properties:**
Assuming a linear, isotropic, and homogeneous (LIH) medium, the electric flux density is directly proportional to the electric field intensity $\mathbf{E}$:

$$\mathbf{D} = \epsilon \mathbf{E} \tag{5}$$

where ϵ is the permittivity of the medium. Substituting this into our enclosed charge equation yields:

$$Q_{\text{enc}} = \int_v \nabla \cdot (\epsilon \mathbf{E})\, dv = \epsilon \int_v (\boldsymbol{\nabla} \cdot E)\, dv \tag{6}$$

4. **Analyzing the Uniform Field:**
   The problem states that the electric field is *uniform*. By definition, a uniform vector field has a constant magnitude and a constant direction at every point in space.

   Let us align the z-axis of our coordinate system with the axis of the cylinder and the direction of the uniform electric field. Thus, the electric field vector can be written as:

$$\mathbf{E} = E_0\hat{\mathbf{z}} \tag{7}$$

   where E_0 is a constant scalar value.

5. **Computing the Divergence:**
   Now we calculate the divergence of this uniform electric field in Cartesian coordinates:

$$\nabla \cdot E = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \tag{8}$$

$$= \frac{\partial(0)}{\partial x} + \frac{\partial(0)}{\partial y} + \frac{\partial(E_0)}{\partial z} \tag{9}$$

   Since the derivative of a constant (E_0) with respect to any spatial variable is exactly zero, we obtain:

$$\nabla \cdot E = 0 + 0 + 0 = 0 \tag{10}$$

6. **Evaluating the Enclosed Charge:**
   Since the divergence of the electric field is zero everywhere inside the cylinder, the volume charge density ρ_v must also be zero everywhere.

$$Q_{\text{enc}} = \epsilon \int_v (0)\, dv = 0 \text{ Coulombs} \tag{11}$$

Final Answer

The total charge enclosed by the unit cylinder placed in a uniform electric field is exactly **0 C**. This implies that the electric flux entering the cylinder perfectly balances the electric flux leaving the cylinder.

1.2 (b) Ring and Disc Charge Superposition (7 Marks)

Problem Statement:
Consider a ring charge of radius 10cm and uniform charge density of +0.5 nC/m and also a disc charge of radius 15cm and uniform charge density σ C/m^2. Both charges are placed in the y-z plane with their centers at the origin. If the electric field intensity at a point of height 20cm lying on the x-axis is same due to the ring and disc charges individually, then find the magnitude of σ. Relative permittivity of the medium is 3.

Detailed Step-by-Step Solution:

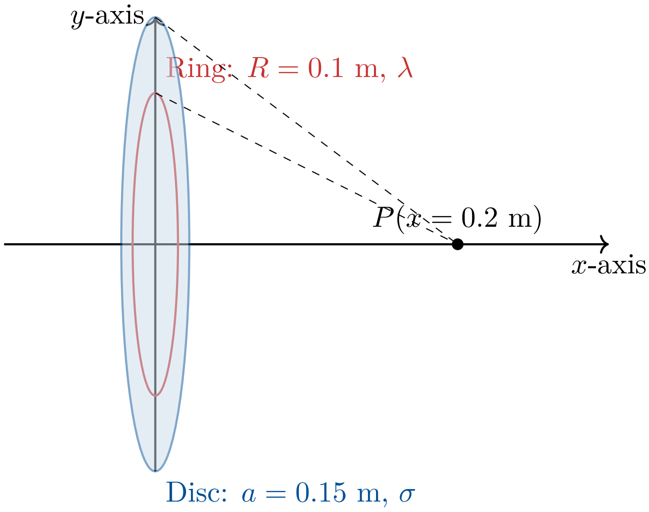


Figure 1: Geometry of the Ring and Disc Charge arrangement.

1. **Identify the Given Parameters:**

   - Permittivity of the medium: $\epsilon = \epsilon_r\epsilon_0 = 3\epsilon_0$
   - Point of interest P coordinate: $x = 20$ cm $= 0.2$ m
   - **Ring Charge Data:**
     - Radius of ring, $R = 10$ cm $= 0.1$ m
     - Linear charge density, $\lambda = 0.5$ nC/m $= 0.5 \times 10^{-9}$ C/m
   - **Disc Charge Data:**
     - Radius of disc, $a = 15$ cm $= 0.15$ m
     - Surface charge density, $\sigma =$?

2. **Calculate Electric Field due to the Ring Charge (E_{ring}):**
   The expression for the electric field intensity along the axis (x-axis) of a uniformly charged circular ring is derived by integrating the differential fields $d\mathbf{E}$ from differential charge elements $dq = \lambda dl$ around the ring. By symmetry, the transverse components cancel out, leaving only the axial component:

$$E_{\text{ring}} = \frac{\lambda \cdot R \cdot x}{2\epsilon(R^2 + x^2)^{3/2}} \tag{12}$$

   Substitute the given numerical values into the ring formula:

$$\begin{aligned} E_{\text{ring}} &= \frac{(0.5 \times 10^{-9}) \cdot (0.1) \cdot (0.2)}{2 \cdot (3\epsilon_0) \cdot (0.1^2 + 0.2^2)^{3/2}} && (13) \\ &= \frac{0.01 \times 10^{-9}}{6\epsilon_0 \cdot (0.01 + 0.04)^{3/2}} && (14) \\ &= \frac{0.01 \times 10^{-9}}{6\epsilon_0 \cdot (0.05)^{3/2}} && (15) \end{aligned}$$

   Calculate the denominator power term:

$$(0.05)^{3/2} = \sqrt{0.05^3} = \sqrt{0.000125} \approx 0.0111803 \tag{16}$$

Continuing the calculation for E_{ring}:

$$E_{\text{ring}} = \frac{0.01 \times 10^{-9}}{6\epsilon_0 \cdot (0.0111803)} \tag{17}$$

$$= \frac{0.01 \times 10^{-9}}{0.06708 \cdot \epsilon_0} \tag{18}$$

$$= \frac{0.149 \times 10^{-9}}{\epsilon_0} \text{ V/m} \quad \text{(Note: leaving in terms of } \epsilon_0 \text{ for easier cancellation later)} \tag{19}$$

Alternative method for keeping epsilon intact: Let's re-calculate using the base ϵ without substituting $3\epsilon_0$ immediately:

$$E_{\text{ring}} = \frac{(0.5 \times 10^{-9})(0.1)(0.2)}{2\epsilon(0.05)^{3/2}} \tag{20}$$

$$= \frac{10^{-11}}{2\epsilon(0.01118)} \tag{21}$$

$$= \frac{0.447 \times 10^{-9}}{\epsilon} \text{ V/m} \tag{22}$$

3. **Calculate Electric Field due to the Disc Charge (E_{disc}):**
The expression for the electric field intensity along the axis of a uniformly charged solid disc can be derived by integrating the fields of infinite concentric ring charges from radius $r = 0$ to $r = a$:

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon}\left[1 - \frac{x}{\sqrt{a^2 + x^2}}\right] \tag{23}$$

Substitute the given numerical values for the disc:

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon}\left[1 - \frac{0.2}{\sqrt{0.15^2 + 0.2^2}}\right] \tag{24}$$

$$= \frac{\sigma}{2\epsilon}\left[1 - \frac{0.2}{\sqrt{0.0225 + 0.04}}\right] \tag{25}$$

$$= \frac{\sigma}{2\epsilon}\left[1 - \frac{0.2}{\sqrt{0.0625}}\right] \tag{26}$$

Since $\sqrt{0.0625} = 0.25$:

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon}\left[1 - \frac{0.2}{0.25}\right] \tag{27}$$

$$= \frac{\sigma}{2\epsilon}[1 - 0.8] \tag{28}$$

$$= \frac{\sigma}{2\epsilon}[0.2] \tag{29}$$

$$= \frac{0.1\sigma}{\epsilon} \text{ V/m} \tag{30}$$

4. **Equate the Fields to Solve for σ:**
The problem states that the electric field intensity due to the ring is exactly equal to the electric field intensity due to the disc at $x = 0.2$ m.

$$E_{\text{ring}} = E_{\text{disc}} \tag{31}$$

$$\frac{0.447 \times 10^{-9}}{\epsilon} = \frac{0.1\sigma}{\epsilon} \tag{32}$$

Notice that the permittivity of the medium (ϵ) appears in the denominator on both sides and cancels out perfectly. This implies the relative permittivity value (3) was effectively a distractor for finding σ under equality conditions!

$$0.447 \times 10^{-9} = 0.1\sigma \tag{33}$$

$$\sigma = \frac{0.447 \times 10^{-9}}{0.1} \tag{34}$$

$$\sigma = 4.47 \times 10^{-9}\ \mathrm{C/m^2} \tag{35}$$

Final Answer

The magnitude of the uniform surface charge density on the disc must be:
$\sigma = \mathbf{4.47 \times 10^{-9}}\ \mathrm{C/m^2}$ (or $4.47\ \mathrm{nC/m^2}$)

1.3 (c) Negative Sign in Potential Difference Equation (4 Marks)

Problem Statement:
Consider that a unit positive charge is moved from point 1 to point 2 by a small distance dl in an electric field. Then why the negative sign is incorporated in the following equation for the potential difference: $\phi_2 - \phi_1 = -\int \mathbf{E} \cdot d\mathbf{l}$?

Detailed Theoretical Explanation:
The fundamental definition of absolute electric potential, and thereby potential difference, is rooted in the concept of **work done by an external agent**. Let us break down the physical mechanics:

1. **Electrostatic Force on the Charge:**
Suppose an electric field $\mathbf{E}$ exists in a region. If a test charge q is placed in this field, it experiences an electrostatic force exerted *by the field* itself. According to Coulomb's definitions, this force is:

$$\mathbf{F}_{\text{electric}} = q\mathbf{E} \tag{36}$$

For a *unit positive charge* ($q = +1$ C), this force is simply equal to the electric field vector:

$$\mathbf{F}_{\text{electric}} = \mathbf{E} \tag{37}$$

2. **Requirement for Quasi-Static Movement:**
To define potential strictly as a function of position (potential energy), we must move the charge from point 1 to point 2 **without imparting any kinetic energy** to it. This means the charge must be moved with zero net acceleration (a quasi-static process).

3. **The External Agent's Role:**
By Newton's Second Law ($\mathbf{F}_{\text{net}} = m\mathbf{a}$), for the acceleration to be zero, the net force on

the test charge must be zero. Therefore, an external force must be applied that perfectly opposes and cancels out the electrostatic force at every instant along the path:

$$\mathbf{F}_{\text{net}} = \mathbf{F}_{\text{external}} + \mathbf{F}_{\text{electric}} = 0 \tag{38}$$
$$\implies \mathbf{F}_{\text{external}} = -\mathbf{F}_{\text{electric}} \tag{39}$$
$$\implies \mathbf{F}_{\text{external}} = -q\mathbf{E} \tag{40}$$

For our unit positive charge, $\mathbf{F}_{\text{external}} = -\mathbf{E}$.

4. **Calculating the Work Done:**
The electrical potential difference between point 2 and point 1, denoted as V_{21} or $(\phi_2 - \phi_1)$, is explicitly defined as the total mechanical work W done **by this external agent** in moving the unit charge along the differential path $d\mathbf{l}$.

The differential work done dW is the dot product of the applied external force and the differential displacement vector:

$$dW = \mathbf{F}_{\text{external}} \cdot d\mathbf{l} \tag{41}$$
$$dW = (-\mathbf{E}) \cdot d\mathbf{l} \tag{42}$$
$$dW = -\mathbf{E} \cdot d\mathbf{l} \tag{43}$$

5. **Total Integral for Potential Difference:**
Integrating this differential work over the entire path from starting point 1 to ending point 2 gives the total potential difference:

$$\phi_2 - \phi_1 = \int_1^2 dW = \int_1^2 (-\mathbf{E} \cdot d\mathbf{l}) = -\int_1^2 \mathbf{E} \cdot d\mathbf{l} \tag{44}$$

Final Answer

Conclusion:
The negative sign is incorporated because potential difference is defined as the work done by an **external force** against the electric field. To move a charge without acceleration, the external force must be exactly opposite in direction to the electric field ($\mathbf{F}_{\text{ext}} = -\mathbf{E}$). The negative sign ensures that moving a positive charge *against* the field lines results in a positive increase in its potential energy.

2 Question 2 - Vector Calculus in Electromagnetics

2.1 (a) Physical Interpretation of Divergence (4 Marks)

Problem Statement:
Correct or justify: "The divergence of a vector field at any location is the total flux of that vector field coming out per unit volume at that given location."

Detailed Justification:
The statement provided is **Correct**.

Proof. In vector calculus, the mathematical definition of the divergence of a vector field $\mathbf{A}$ at a specific point P is formally given by taking the limit of the surface integral (net flux) over a closed surface S bounding a volume Δv as the volume shrinks to zero around point P:

$$\boldsymbol{\nabla} \cdot A = \lim_{\Delta v \to 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{s}}{\Delta v} \tag{45}$$

Let us dissect the terms in this equation to map them to the English statement:

- $\oint_S \mathbf{A} \cdot d\mathbf{s}$: This is the surface integral of the vector field over a closed boundary. Physically, this represents the **net outward flux** of the vector field $\mathbf{A}$ passing through the closed surface S.
- $\frac{1}{\Delta v}$: By dividing the net outward flux by the volume Δv that the surface encloses, we obtain the **flux per unit volume**.
- $\lim_{\Delta v \to 0}$: By shrinking the volume to an infinitesimally small element centered at a specific location, we evaluate the property exactly at **that given location** (a point property, not a bulk property).

Therefore, the divergence mathematically measures exactly what the sentence describes: the rate at which "flux" is generated (or consumed) per unit volume at a single, precise geometric coordinate. A positive divergence indicates a localized "source" of flux, a negative divergence indicates a localized "sink", and a zero divergence indicates that whatever flux enters the infinitesimal volume also perfectly exits it (a solenoidal field). □

2.2 (b) Derivation of Divergence in Cylindrical Coordinates (7 Marks)

Problem Statement:
Derive an expression for the divergence of a vector in cylindrical coordinate system.

Detailed Step-by-Step Derivation:
To derive the divergence formula in a specific coordinate system from scratch, we use the fundamental definition of divergence applied to an infinitesimal volume element native to that coordinate system.

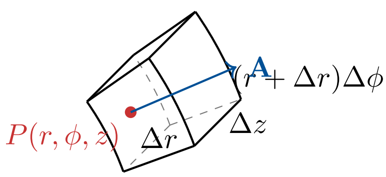


Figure 2: Differential volume element in Cylindrical Coordinates (r, ϕ, z).

Let an arbitrary vector field be expressed in cylindrical coordinates as:

$$\mathbf{A} = A_r\hat{\mathbf{r}} + A_\phi\hat{\phi} + A_z\hat{\mathbf{z}} \tag{46}$$

Consider an infinitesimal volume element Δv at point $P(r, \phi, z)$ defined by differential increments Δr, $\Delta\phi$, and Δz. The geometric dimensions of this curved "brick" are:

- Length along radial direction $= \Delta r$
- Length along azimuthal (arc) direction $= r\Delta\phi$ (at inner face) and $(r + \Delta r)\Delta\phi$ (at outer face)
- Length along axial direction $= \Delta z$

The total enclosed differential volume is:

$$\Delta v = r\,\Delta r\,\Delta\phi\,\Delta z \tag{47}$$

We calculate the net outward flux $\Phi = \oint \mathbf{A} \cdot d\mathbf{s}$ across all 6 faces (3 pairs).

Step 1: Flux through the Front and Back faces (Radial direction, $\hat{\mathbf{r}}$)

- The inner face (back) is located at radius r. Its area is $\Delta S_{\text{inner}} = r\Delta\phi\Delta z$. The flux entering this face is roughly $A_r(r) \cdot r\Delta\phi\Delta z$.
- The outer face (front) is located at radius $(r + \Delta r)$. Its area is larger: $\Delta S_{\text{outer}} = (r + \Delta r)\Delta\phi\Delta z$. The flux leaving this face is $A_r(r + \Delta r) \cdot (r + \Delta r)\Delta\phi\Delta z$.

The net outward flux in the radial direction (Φ_r) is the difference:

$$\Phi_r = \text{Flux out} - \text{Flux in} \tag{48}$$

$$\Phi_r = [A_r(r + \Delta r)(r + \Delta r) - A_r(r)r]\,\Delta\phi\Delta z \tag{49}$$

Using the first-order Taylor series expansion (or the formal definition of a partial derivative as $\Delta r \to 0$):

$$\frac{f(r + \Delta r) - f(r)}{\Delta r} \approx \frac{\partial f}{\partial r} \tag{50}$$

Let $f(r) = rA_r$. Then:

$$[(r + \Delta r)A_r(r + \Delta r) - rA_r(r)] \approx \frac{\partial}{\partial r}(rA_r)\Delta r \tag{51}$$

Substituting this back:

$$\Phi_r = \frac{\partial}{\partial r}(rA_r)\Delta r\Delta\phi\Delta z \tag{52}$$

Step 2: Flux through the Left and Right faces (Azimuthal direction, $\hat{\phi}$)

- These faces are located at angle ϕ and $\phi + \Delta\phi$.
- The surface area for both faces is exactly the same: $\Delta S_\phi = \Delta r\Delta z$.
- Flux entering at ϕ is $A_\phi(\phi)\Delta r\Delta z$.
- Flux leaving at $\phi + \Delta\phi$ is $A_\phi(\phi + \Delta\phi)\Delta r\Delta z$.

The net outward flux in the azimuthal direction (Φ_ϕ) is:

$$\begin{aligned} \Phi_\phi &= [A_\phi(\phi + \Delta\phi) - A_\phi(\phi)]\,\Delta r \Delta z && (53) \\ &\approx \left(\frac{\partial A_\phi}{\partial \phi}\Delta\phi\right)\Delta r \Delta z && (54) \\ \Phi_\phi &= \frac{\partial A_\phi}{\partial \phi}\Delta r \Delta\phi \Delta z && (55) \end{aligned}$$

Step 3: Flux through the Top and Bottom faces (Axial direction, $\hat{\mathbf{z}}$)

- These faces are located at height z and $z + \Delta z$.
- The surface area for both faces is identical: $\Delta S_z = r\Delta r \Delta\phi$.
- Flux entering at z is $A_z(z) r\Delta r\Delta\phi$.
- Flux leaving at $z + \Delta z$ is $A_z(z + \Delta z) r\Delta r\Delta\phi$.

The net outward flux in the axial direction (Φ_z) is:

$$\begin{aligned} \Phi_z &= [A_z(z + \Delta z) - A_z(z)]\, r\Delta r \Delta\phi && (56) \\ &\approx \left(\frac{\partial A_z}{\partial z}\Delta z\right) r\Delta r \Delta\phi && (57) \\ \Phi_z &= \frac{\partial A_z}{\partial z} r\Delta r \Delta\phi \Delta z && (58) \end{aligned}$$

Step 4: Final Divergence Assembly By the definition of divergence, we sum the total flux and divide by the volume $\Delta v = r\Delta r\Delta\phi\Delta z$:

$$\nabla \cdot A = \lim_{\Delta v \to 0} \frac{\Phi_r + \Phi_\phi + \Phi_z}{\Delta v} \tag{59}$$

$$\nabla \cdot A = \frac{\frac{\partial}{\partial r}(rA_r)\Delta r\Delta\phi\Delta z + \frac{\partial A_\phi}{\partial \phi}\Delta r\Delta\phi\Delta z + \frac{\partial A_z}{\partial z} r\Delta r\Delta\phi\Delta z}{r\Delta r\Delta\phi\Delta z} \tag{60}$$

Dividing term by term, the differential spatial lengths cancel out entirely:

$$\nabla \cdot A = \frac{\frac{\partial}{\partial r}(rA_r)}{r} + \frac{\frac{\partial A_\phi}{\partial \phi}}{r} + \frac{\frac{\partial A_z}{\partial z} r}{r} \tag{61}$$

Final Answer

Final Expression for Divergence in Cylindrical Coordinates:

$$\nabla \cdot A = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \tag{62}$$

2.3 (c) Workless Displacement in a Uniform Field (5 Marks)

Problem Statement:
A uniform electric field is parallel to the z-axis. In which direction can a unit positive charge be displaced by a distance 'd' without any external work being done?

Detailed Solution:

1. **Define the Electric Field:**
The problem states the uniform electric field is parallel to the z-axis. We can write this vector field as:
$$\mathbf{E} = E_0\hat{\mathbf{k}} \tag{63}$$
where E_0 is the constant magnitude of the electric field and $\hat{\mathbf{k}}$ (or $\hat{\mathbf{z}}$) is the unit vector in the z-direction.

2. **Define the Work Done Equation:**
As derived in Question 1(c), the external work W done in displacing a unit charge ($q = 1$ C) by an arbitrary differential distance vector $d\mathbf{l}$ is given by:
$$W = -\int \mathbf{E} \cdot d\mathbf{l} \tag{64}$$
Let the total displacement vector over a straight line distance d be:
$$\mathbf{d} = d_x\hat{\mathbf{i}} + d_y\hat{\mathbf{j}} + d_z\hat{\mathbf{k}} \tag{65}$$
Since the field is uniform, the integral evaluates directly to a simple dot product:
$$W = -\mathbf{E} \cdot \mathbf{d} \tag{66}$$

3. **Impose the Zero Work Condition:**
The problem asks for the condition where **no external work is done**, which translates mathematically to setting $W = 0$:
$$W = -(E_0\hat{\mathbf{k}}) \cdot (d_x\hat{\mathbf{i}} + d_y\hat{\mathbf{j}} + d_z\hat{\mathbf{k}}) = 0 \tag{67}$$
Evaluate the dot product. The dot products of orthogonal unit vectors ($\hat{\mathbf{k}} \cdot \hat{\mathbf{i}}$ and $\hat{\mathbf{k}} \cdot \hat{\mathbf{j}}$) are zero, and $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$:
$$W = -E_0(0 \cdot d_x + 0 \cdot d_y + 1 \cdot d_z) \tag{68}$$
$$W = -E_0 d_z = 0 \tag{69}$$

4. **Analyze the Result:**
Assuming the electric field E_0 is non-zero, the only way for the equation $-E_0 d_z = 0$ to hold true is if:
$$d_z = 0 \tag{70}$$
This means the displacement vector $\mathbf{d}$ must have an absolutely zero component in the z-direction. The displacement must be entirely comprised of x and y components:
$$\mathbf{d} = d_x\hat{\mathbf{i}} + d_y\hat{\mathbf{j}} \tag{71}$$

Final Answer

Conclusion:
A unit positive charge can be displaced in **any direction strictly within the x-y plane** (or any plane that is perfectly perpendicular/normal to the z-axis, defined by $z =$ constant) without any external work being done. In physics terminology, these planes normal to a uniform electric field are defined as *Equipotential Surfaces*, and moving a charge along an equipotential surface inherently requires zero work because the force vector is always perfectly perpendicular to the displacement vector.

3 Question 3 - Boundary Conditions

3.1 (a) Electric Field Just Off vs. On a Conductor Surface (4 Marks)

Problem Statement:
Correct or justify: "Electric field intensity just off the conductor surface is half of the electric field intensity exactly on the conductor surface".

Detailed Justification:
The statement is **Incorrect**. It is precisely reversed.

Proof. Correction: The electric field intensity **just off** (immediately outside) the conductor surface is exactly **double** the electric field intensity acting **exactly on** the conductor surface.

Detailed Physical Proof:
Consider a metallic conductor with a local surface charge density σ.

1. Field Just Outside the Conductor (E_{off}):
By applying Gauss's law to a pillbox straddling the conductor surface, the boundary condition dictates that the electric field immediately adjacent to the outside of a conductor in a dielectric medium is purely normal to the surface and is given by:

$$E_{\text{off}} = \frac{\sigma}{\epsilon} \tag{72}$$

2. Field Exactly On the Conductor Layer (E_{on}):
To find the field driving the charge layer itself, we must use the principle of superposition. We virtually divide the total conductor surface charge into two distinct parts:

- Part A: A very tiny, microscopic circular patch of charge $dq = \sigma ds$ exactly at the point of interest.
- Part B: The entire remainder of the conductor's surface charge.

By Gauss's law for an infinite sheet, the tiny local patch (Part A) generates an electric field pointing outwards on both of its sides with magnitude:

$$E_{\text{patch}} = \frac{\sigma}{2\epsilon} \tag{73}$$

Let the electric field produced by the rest of the conductor (Part B) be E_{rest}.

Just outside the conductor, the total field is the sum of the patch field pointing out and the rest field pointing out:

$$E_{\text{off}} = E_{\text{patch}} + E_{\text{rest}} \implies \frac{\sigma}{\epsilon} = \frac{\sigma}{2\epsilon} + E_{\text{rest}} \tag{74}$$

Solving for E_{rest} yields:

$$E_{\text{rest}} = \frac{\sigma}{\epsilon} - \frac{\sigma}{2\epsilon} = \frac{\sigma}{2\epsilon} \tag{75}$$

Now, consider a physical charge existing exactly **on** the boundary layer (within the patch). A charge cannot exert a net force upon itself (Newton's Third Law internal cancellation). Therefore, the only field capable of exerting an electrical pressure (force) upon the charges in the patch is the field originating from the *rest* of the conductor, E_{rest}.

Consequently, the effective field intensity acting *on* the surface is:

$$E_{\text{on}} = E_{\text{rest}} = \frac{\sigma}{2\epsilon} \tag{76}$$

Conclusion of Proof:
Comparing the two results:

$$E_{\text{on}} = \frac{\sigma}{2\epsilon} \quad \text{and} \quad E_{\text{off}} = \frac{\sigma}{\epsilon} \implies E_{\text{on}} = \frac{1}{2}E_{\text{off}} \tag{77}$$

The field on the surface is half the field just off the surface, contradicting the original statement which claimed the off-surface field was half the on-surface field. □

3.2 (b) Normal Field at Conductor-Dielectric Boundary (4 Marks)

Problem Statement:
Correct or justify: "On any conductor-dielectric boundary... $E_n = \sigma$"

Detailed Justification:
The statement is **Incorrect**.

Proof. Correction: The correct mathematical boundary relation is $E_n = \frac{\sigma}{\epsilon}$, where ϵ is the absolute permittivity of the adjacent dielectric medium.

Detailed Physical Proof:
The primary electrostatic boundary condition for the normal component of fields across any interface states that the discontinuity in the normal component of the electric flux density $\mathbf{D}$ is exactly equal to the free surface charge density ρ_s (often written as σ) residing on that interface:

$$D_{1n} - D_{2n} = \rho_s = \sigma \tag{78}$$

Let Medium 2 be the interior of a perfect electrical conductor under electrostatic equilibrium. Inside a perfect conductor, all internal macroscopic electric fields are strictly zero to prevent infinite currents.

$$\mathbf{E}_2 = 0 \implies \mathbf{D}_2 = \epsilon_2 \mathbf{E}_2 = 0 \implies D_{2n} = 0 \tag{79}$$

Substituting $D_{2n} = 0$ into the boundary condition equation gives:

$$D_{1n} - 0 = \sigma \implies D_{1n} = \sigma \tag{80}$$

This shows that the normal *flux density* (D_n) is equal to σ. However, the statement in the question asserts that the *electric field intensity* (E_n) is equal to σ.

To find E_n, we must use the constitutive relation for a linear isotropic dielectric medium (Medium 1), which is $\mathbf{D}_1 = \epsilon_1 \mathbf{E}_1$, where ϵ_1 is the permittivity ($\epsilon_0 \epsilon_r$). Taking the normal component:

$$D_{1n} = \epsilon_1 E_{1n} \tag{81}$$

Substituting this into our previous derived equation $D_{1n} = \sigma$:

$$\epsilon_1 E_{1n} = \sigma \implies E_{1n} = \frac{\sigma}{\epsilon_1} \tag{82}$$

The statement $E_n = \sigma$ is fundamentally flawed because it completely ignores the material property (permittivity, ϵ) of the dielectric medium and also violates dimensional analysis (V/m does not equal C/m^2). □

3.3 (c) Universal Validity of Normal Flux Boundary Condition (8 Marks)

Problem Statement:
Prove that the equation $D_{2n} - D_{1n} = \rho_s$ is valid for dielectric-dielectric boundary as well as for conductor-dielectric boundary.

Detailed Proof and Derivation:
To prove universal validity, we must first derive the equation from first principles without assuming any specific material properties, and then apply it to the two distinct cases to show it natively handles both without contradiction.

General Derivation using a Gaussian Pillbox:

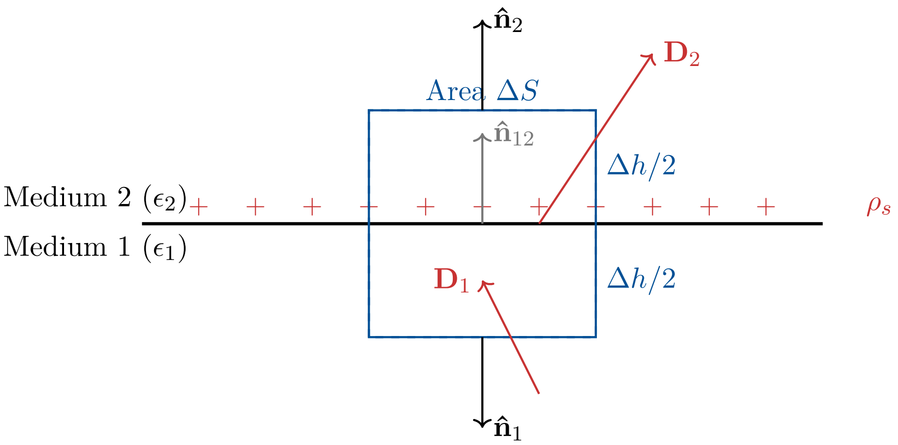


Figure 3: Gaussian Pillbox spanning a boundary carrying surface charge ρ_s.

Let a boundary separating Medium 1 and Medium 2 lie in an arbitrary plane. We construct a small Gaussian "pillbox" (a shallow cylinder) of base cross-sectional area ΔS and total height Δh positioned symmetrically such that it is bisected by the boundary surface.

Apply Gauss's Law in integral form to the pillbox:

$$\oint_{\text{pillbox}} \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enc}} \tag{83}$$

The closed surface integral can be broken into three parts: the top face (in Medium 2), the bottom face (in Medium 1), and the cylindrical side wall:

$$\int_{\text{top}} \mathbf{D}_2 \cdot d\mathbf{s}_2 + \int_{\text{bottom}} \mathbf{D}_1 \cdot d\mathbf{s}_1 + \int_{\text{sides}} \mathbf{D} \cdot d\mathbf{s}_{\text{side}} = Q_{\text{enc}} \tag{84}$$

We now take the limit as the height of the pillbox shrinks to zero ($\Delta h \to 0$), keeping the area ΔS constant. As $\Delta h \to 0$, the surface area of the side walls approaches zero, forcing the flux contribution from the sides to vanish completely:

$$\lim_{\Delta h \to 0} \int_{\text{sides}} \mathbf{D} \cdot d\mathbf{s}_{\text{side}} = 0 \tag{85}$$

Assuming ΔS is small enough that $\mathbf{D}$ is uniform across it, the top and bottom integrals simplify to dot products:

$$\mathbf{D}_2 \cdot (\hat{\mathbf{n}}_2 \Delta S) + \mathbf{D}_1 \cdot (\hat{\mathbf{n}}_1 \Delta S) = Q_{\text{enc}} \tag{86}$$

Let us define a single unit normal vector $\hat{\mathbf{n}}$ pointing from Medium 1 into Medium 2. Therefore, the outward normal for the top face is $\hat{\mathbf{n}}_2 = \hat{\mathbf{n}}$, and the outward normal for the bottom face is pointing opposite, $\hat{\mathbf{n}}_1 = -\hat{\mathbf{n}}$. The enclosed charge Q_{enc} captured by the flat pillbox cross-section is the free surface charge density ρ_s multiplied by the area ΔS:

$$Q_{\text{enc}} = \rho_s \Delta S \tag{87}$$

Substituting these into our Gauss's law equation:

$$\mathbf{D}_2 \cdot (\hat{\mathbf{n}}\Delta S) + \mathbf{D}_1 \cdot (-\hat{\mathbf{n}}\Delta S) = \rho_s \Delta S \tag{88}$$
$$(\mathbf{D}_2 \cdot \hat{\mathbf{n}})\Delta S - (\mathbf{D}_1 \cdot \hat{\mathbf{n}})\Delta S = \rho_s \Delta S \tag{89}$$

Dividing through by the non-zero common area scalar ΔS, and recognizing that the dot product with the normal vector $\hat{\mathbf{n}}$ simply extracts the normal component scalar (D_n), we arrive at the general root equation:

$$D_{2n} - D_{1n} = \rho_s \tag{90}$$

This derivation made zero assumptions about the physical state of the media, relying solely on geometric topology and Maxwell's equations. Now we test its validity against the two specific cases.

Case 1: Dielectric-Dielectric Boundary
When separating two ideal dielectric media (insulators), there are typically no highly mobile free electrons available to form a persistent surface charge layer. In the vast majority of practical dielectric-dielectric interfaces, the free surface charge is identically zero ($\rho_s = 0$). Substituting $\rho_s = 0$ into our general equation yields:

$$D_{2n} - D_{1n} = 0 \implies D_{2n} = D_{1n} \tag{91}$$

This is the universally taught boundary condition for contiguous dielectrics (normal component of D is continuous). *However*, the general equation $D_{2n} - D_{1n} = \rho_s$ remains strictly mathematically valid even here. If a physicist deliberately sprays a layer of free static charge onto the interface between two glass blocks, then $\rho_s \neq 0$, and the full generalized equation must be used. Therefore, it is perfectly valid for dielectric-dielectric boundaries.

Case 2: Conductor-Dielectric Boundary
Let Medium 1 be a perfect electrical conductor and Medium 2 be a dielectric. Under static conditions, no internal electric field can exist inside a perfect conductor. If a field did exist, it would immediately move infinite charge until the internal field was canceled.

$$\mathbf{E}_1 = 0 \implies \mathbf{D}_1 = \epsilon_1 \mathbf{E}_1 = 0 \tag{92}$$

Because the total flux density vector inside is zero, its normal component is also trivially zero: $D_{1n} = 0$. Substituting this hard physical constraint into our generalized equation yields:

$$D_{2n} - (0) = \rho_s \tag{93}$$
$$D_{2n} = \rho_s \tag{94}$$

This resulting expression ($D_{2n} = \rho_s$) is the exact, standard boundary condition explicitly taught for conductor-dielectric interfaces.

Final Answer

Conclusion:
Because the equation $D_{2n} - D_{1n} = \rho_s$ is derived from the integral form of Gauss's Law without imposing any material-specific constraints, and because it algebraically reduces

to the correct phenomenological forms when the specific constraints of dielectrics ($\rho_s = 0$) or conductors ($D_1 = 0$) are applied, it is proven to be the **universally valid master equation** for the normal component of flux density across any electromagnetic boundary.

Field Theory Solution
Question 4(a), 4(b), 4(c)

4(a) Orthogonal coordinate system with one constant distance surface and two constant angle surfaces

The required coordinate system is the **spherical coordinate system**.

In spherical coordinates, a point P is represented by

$$P(r, \theta, \phi)$$

where

$$r = \text{radial distance from origin,}$$

$$\theta = \text{polar angle measured from the positive } z\text{-axis,}$$

$$\phi = \text{azimuthal angle measured in the } xy\text{-plane from the positive } x\text{-axis.}$$

Diagram of spherical coordinate system

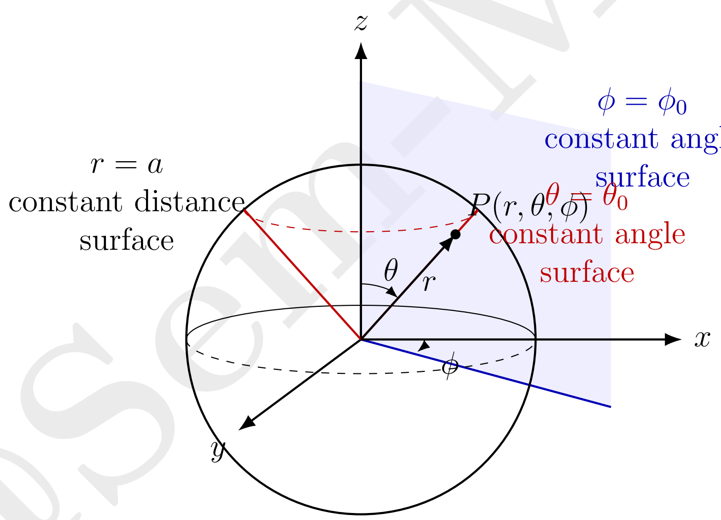


Figure 1: Spherical coordinate system: $r =$ constant is a sphere, $\theta =$ constant is a cone, and $\phi =$ constant is a half-plane.

Constant coordinate surfaces

In spherical coordinates, the three constant coordinate surfaces are:

$$\boxed{r = \text{constant}}$$

This represents a **sphere**. Every point on this surface is at the same distance from the origin. Therefore, it is called a **constant distance surface**.

$$\boxed{\theta = \text{constant}}$$

This represents a **right circular cone** with its vertex at the origin and its axis along the z-axis. Since θ is an angle, this is a **constant angle surface**.

$$\boxed{\phi = \text{constant}}$$

This represents a **half-plane** passing through the z-axis. Since ϕ is also an angle, this is another **constant angle surface**.

Orthogonality

The spherical coordinate system is an **orthogonal coordinate system** because the three unit vectors

$$\hat{a}_r, \qquad \hat{a}_\theta, \qquad \hat{a}_\phi$$

are mutually perpendicular to each other.

Therefore,

$$\hat{a}_r \cdot \hat{a}_\theta = 0,$$
$$\hat{a}_\theta \cdot \hat{a}_\phi = 0,$$
$$\hat{a}_\phi \cdot \hat{a}_r = 0.$$

The differential length element in spherical coordinates is

$$d\vec{l} = \hat{a}_r\, dr + \hat{a}_\theta\, r\, d\theta + \hat{a}_\phi\, r \sin\theta\, d\phi.$$

Hence, the scale factors are

$$h_r = 1, \qquad h_\theta = r, \qquad h_\phi = r\sin\theta.$$

$$\boxed{\text{The required coordinate system is the spherical coordinate system.}}$$

It has one constant distance surface:

$$\boxed{r = \text{constant}}$$

and two constant angle surfaces:

$$\boxed{\theta = \text{constant}}, \qquad \boxed{\phi = \text{constant}}.$$

4(b) Derivation of FDM equations for unknown node potentials in a 2-D system with equal nodal distances in homogeneous medium

In a homogeneous medium, the permittivity ε is constant. For a charge-free electrostatic region, the potential V satisfies Laplace's equation:

$$\nabla^2 V = 0.$$

In two dimensions,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0.$$

Finite difference grid

Let the node whose potential is unknown be P. Its neighbouring nodes are east, west, north, and south.

$$V_P = V_{i,j}$$

$$V_E = V_{i+1,j}, \qquad V_W = V_{i-1,j},$$

$$V_N = V_{i,j+1}, \qquad V_S = V_{i,j-1}.$$

The nodal spacing in both x and y directions is equal to h.

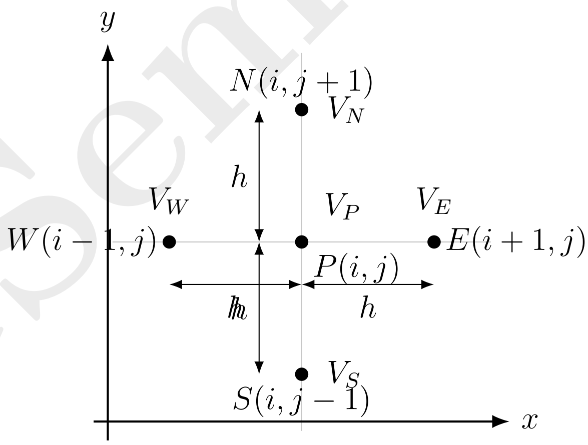


Figure 2: Five-point finite difference molecule for a two-dimensional node.

Step 1: Start from Laplace's equation

For a homogeneous charge-free medium,

$$\nabla^2 V = 0.$$

In two dimensions,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0.$$

Step 2: Approximate the second derivative in the x-direction

Using the central difference formula,

$$\left(\frac{\partial^2 V}{\partial x^2}\right)_{i,j} \approx \frac{V_{i+1,j} - 2V_{i,j} + V_{i-1,j}}{h^2}.$$

Therefore,

$$\left(\frac{\partial^2 V}{\partial x^2}\right)_{P} \approx \frac{V_E - 2V_P + V_W}{h^2}.$$

Step 3: Approximate the second derivative in the y-direction

Similarly,

$$\left(\frac{\partial^2 V}{\partial y^2}\right)_{i,j} \approx \frac{V_{i,j+1} - 2V_{i,j} + V_{i,j-1}}{h^2}.$$

Therefore,

$$\left(\frac{\partial^2 V}{\partial y^2}\right)_{P} \approx \frac{V_N - 2V_P + V_S}{h^2}.$$

Step 4: Substitute in Laplace's equation

Now,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0.$$

Substituting the finite difference approximations,

$$\frac{V_E - 2V_P + V_W}{h^2} + \frac{V_N - 2V_P + V_S}{h^2} = 0.$$

Since both terms have the same denominator h^2,

$$V_E - 2V_P + V_W + V_N - 2V_P + V_S = 0.$$

Therefore,

$$V_E + V_W + V_N + V_S - 4V_P = 0.$$

Hence,

$$4V_P = V_E + V_W + V_N + V_S.$$

So,

$$\boxed{V_P = \frac{V_E + V_W + V_N + V_S}{4}}$$

or in indexed form,

$$\boxed{V_{i,j} = \frac{1}{4}\left[V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}\right]}$$

Final finite difference equation

The finite difference equation for an interior node in a homogeneous charge-free 2-D region is

$$\boxed{4V_{i,j} - V_{i+1,j} - V_{i-1,j} - V_{i,j+1} - V_{i,j-1} = 0}$$

This is known as the **five-point finite difference formula** or **five-point molecule**. The physical meaning of the result is very important:

The potential at an interior node is the average of the potentials at its four neighbouring nodes.

Thus,

$$\boxed{V_P = \frac{V_E + V_W + V_N + V_S}{4}}$$

General form if space charge density is present

If charge density ρ_v is present, then Poisson's equation is used:

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon}.$$

Using the same finite difference method,

$$\frac{V_E + V_W + V_N + V_S - 4V_P}{h^2} = -\frac{\rho_v}{\varepsilon}.$$

Therefore,

$$4V_P - V_E - V_W - V_N - V_S = \frac{\rho_v h^2}{\varepsilon}.$$

Hence,

$$\boxed{V_P = \frac{V_E + V_W + V_N + V_S}{4} + \frac{\rho_v h^2}{4\varepsilon}}$$

For a charge-free region, $\rho_v = 0$, and this reduces to Laplace's equation.

4(c) Correction or justification of the statement

"The relative permittivity of the liquid dielectric used for impregnation must be much higher compared to the relative permittivity of the solid insulation being impregnated."

The statement is **not correct as written**.

The liquid dielectric used for impregnation should not necessarily have relative permittivity much higher than that of the solid insulation. Rather, its relative permittivity should be **close to that of the solid insulation**, so that electric stress is distributed uniformly and no part of the insulation is overstressed.

Reason using dielectric boundary condition

Consider a composite dielectric system having a solid insulation and a liquid dielectric layer between two electrodes.

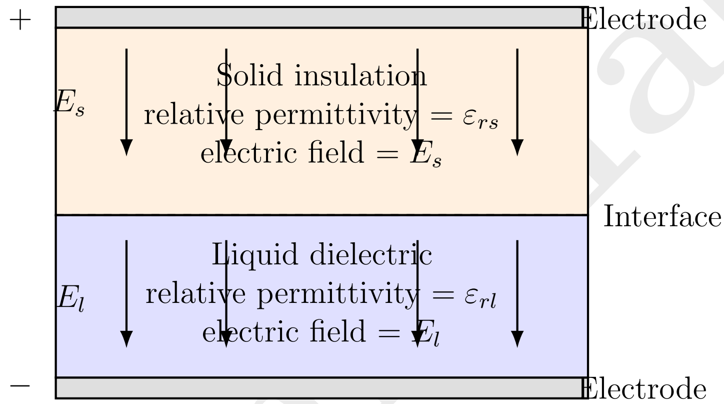


Figure 3: Electric field distribution in a composite solid-liquid dielectric system.

At the interface between two dielectrics, the normal component of electric flux density D is continuous if there is no free surface charge.

Therefore,

$$D_s = D_l.$$

Now,

$$D_s = \varepsilon_0 \varepsilon_{rs} E_s$$

and

$$D_l = \varepsilon_0 \varepsilon_{rl} E_l.$$

Since,

$$D_s = D_l,$$

we get

$$\varepsilon_0 \varepsilon_{rs} E_s = \varepsilon_0 \varepsilon_{rl} E_l.$$

Cancelling ε_0,

$$\varepsilon_{rs} E_s = \varepsilon_{rl} E_l.$$

Therefore,

$$\boxed{\frac{E_s}{E_l} = \frac{\varepsilon_{rl}}{\varepsilon_{rs}}}$$

This shows that electric field intensity is inversely related to permittivity.

Effect of very high liquid permittivity

If the liquid dielectric has much higher relative permittivity than the solid insulation, then

$$\varepsilon_{rl} \gg \varepsilon_{rs}.$$

From the relation,

$$\frac{E_s}{E_l} = \frac{\varepsilon_{rl}}{\varepsilon_{rs}},$$

we get

$$E_s \gg E_l.$$

This means that most of the electric stress will come on the solid insulation.
So, the solid insulation may become overstressed, which is not desirable.

Correct requirement

The liquid dielectric used for impregnation should have:

1. relative permittivity close to that of the solid insulation,

2. high dielectric strength,

3. low dielectric loss,

4. good chemical stability,

5. good thermal stability,

6. ability to fill voids and pores inside the solid insulation,

7. no harmful chemical reaction with the solid insulation.

Why impregnation is done

Solid insulation may contain small air voids or pores. Air has very low relative permittivity:

$$\varepsilon_{r,\text{air}} \approx 1.$$

If air voids remain inside the insulation, electric field becomes very high inside the voids because air has low permittivity and low dielectric strength. This may cause partial discharge and breakdown.

Therefore, impregnation is done to replace air voids by liquid dielectric.

The purpose is not to make the liquid permittivity much higher than the solid permittivity. The purpose is to remove air voids and improve the dielectric strength of the insulation system.

Corrected statement

The corrected statement is:

$$\boxed{\text{The liquid dielectric used for impregnation should have relative permittivity close to that of the solid}}$$

It should be higher than air and should fill the voids properly, but a very large difference between the permittivity of liquid and solid insulation is undesirable because it causes non-uniform electric stress distribution.

Final conclusion

$$\boxed{\text{The given statement is incorrect.}}$$

The relative permittivity of the liquid dielectric should be properly matched with that of the solid insulation. If the liquid permittivity is much higher than the solid permittivity, the solid insulation may experience excessive electric stress. Hence, for good insulation performance, the liquid dielectric should have permittivity close to the solid insulation and should also possess high dielectric strength and low dielectric loss.

5 Question 5 - Equipotentials, Vector Fields, and Cables

5.1 (a) Equipotential Cylinders of Line Charge and Image (8 Marks)

Problem Statement:
Prove that the equipotentials due to an infinitely long line charge and its image wrt an infinitely long conducting plane are cylinders.

Detailed Step-by-Step Proof:

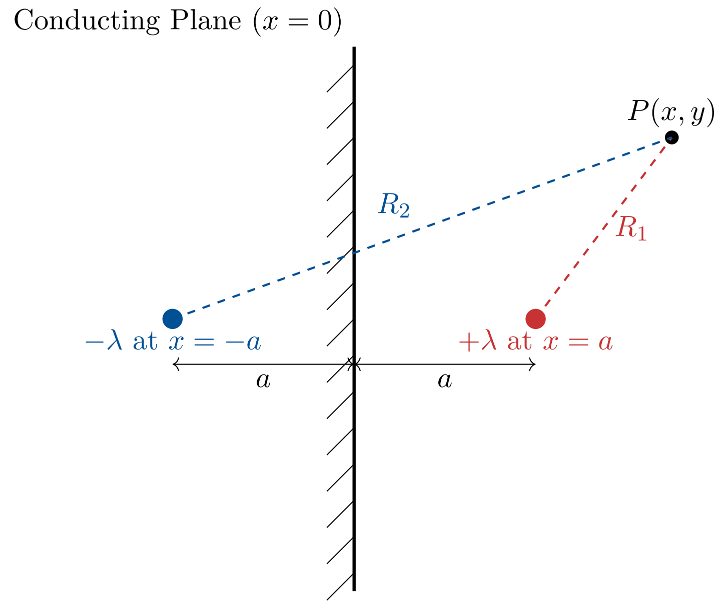


Figure 5: Method of Images for a Line Charge next to a conducting plane.

1. **Setup using Method of Images:**
Consider an infinitely long line charge with uniform linear charge density $+\lambda$ (C/m) placed parallel to an infinite, grounded conducting plane at a distance a. Let the plane lie at $x = 0$. Thus, the line charge is at $x = a$.

   According to the Method of Images, the electric field and potential in the region $x > 0$ are identical to a free-space system where the conducting plane is removed and replaced by a fictitious "image" line charge of $-\lambda$ placed symmetrically at $x = -a$.

2. **Potential of a Single Line Charge:**
The absolute electric potential V at a radial distance r from a single infinitely long line charge λ is derived by integrating the electric field $E = \lambda/(2\pi\epsilon r)$:

$$V(r) = -\int E dr = -\frac{\lambda}{2\pi\epsilon}\ln(r) + C \tag{108}$$

   where C is a constant of integration.

3. **Superposition for the Two-Wire System:**
Applying the principle of superposition, the total potential V_P at an arbitrary spatial point $P(x, y)$ is the algebraic sum of the potentials from the positive charge and the negative image charge. (Since the system extends to infinity in z, the problem reduces to a 2D

cross-sectional analysis in the $x - y$ plane).

$$V_P = V_+ + V_- \tag{109}$$

$$V_P = \left(-\frac{\lambda}{2\pi\epsilon}\ln(R_1) + C_1\right) + \left(-\frac{-\lambda}{2\pi\epsilon}\ln(R_2) + C_2\right) \tag{110}$$

where R_1 is the distance from $+\lambda$ to P, and R_2 is the distance from $-\lambda$ to P. By setting the potential to exactly zero midway between them (at the grounded plane $x = 0$ where $R_1 = R_2$), the constants $C_1 + C_2$ must sum to zero. Thus:

$$V_P = -\frac{\lambda}{2\pi\epsilon}\ln(R_1) + \frac{\lambda}{2\pi\epsilon}\ln(R_2) \tag{111}$$

$$V_P = \frac{\lambda}{2\pi\epsilon}\left[\ln(R_2) - \ln(R_1)\right] \tag{112}$$

$$V_P = \frac{\lambda}{2\pi\epsilon}\ln\left(\frac{R_2}{R_1}\right) \tag{113}$$

4. **Express Distances in Cartesian Coordinates:**
   Using the distance formula in 2D, the radial distances to point $P(x, y)$ are:

$$R_1 = \sqrt{(x-a)^2 + y^2} \quad \text{(Distance from } x = a\text{)} \tag{114}$$

$$R_2 = \sqrt{(x+a)^2 + y^2} \quad \text{(Distance from } x = -a\text{)} \tag{115}$$

5. **Define the Equipotential Condition:**
   An equipotential surface is defined as a locus of points where the potential is identically equal to some constant value V_0.

$$\frac{\lambda}{2\pi\epsilon}\ln\left(\frac{R_2}{R_1}\right) = V_0 \tag{116}$$

   Rearranging to isolate the distance ratio:

$$\ln\left(\frac{R_2}{R_1}\right) = \frac{2\pi\epsilon V_0}{\lambda} \tag{117}$$

$$\frac{R_2}{R_1} = e^{\frac{2\pi\epsilon V_0}{\lambda}} \tag{118}$$

   Since π, ϵ, V_0, and λ are all constants, the entire right side is a dimensionless constant. Let's call it K.

$$\frac{R_2}{R_1} = K \implies R_2 = KR_1 \tag{119}$$

6. **Algebraic Expansion into Geometric Form:**
   Square both sides to remove the square roots:

$$R_2^2 = K^2 R_1^2 \tag{120}$$

$$(x+a)^2 + y^2 = K^2\left[(x-a)^2 + y^2\right] \tag{121}$$

   Expand the binomials:

$$x^2 + 2ax + a^2 + y^2 = K^2(x^2 - 2ax + a^2 + y^2) \tag{122}$$

   Bring all terms to the left side and group them by powers of x and y:

$$x^2 - K^2x^2 + y^2 - K^2y^2 + 2ax + 2aK^2x + a^2 - K^2a^2 = 0 \tag{123}$$

$$x^2(1-K^2) + y^2(1-K^2) + 2ax(1+K^2) + a^2(1-K^2) = 0 \tag{124}$$

Divide the entire polynomial equation by the coefficient of the squared terms $(1 - K^2)$:

$$x^2 + y^2 + 2ax\left(\frac{1+K^2}{1-K^2}\right) + a^2 = 0 \tag{125}$$

This is now in the general form of a conic section $x^2 + y^2 + Dx + Ey + F = 0$. Since the coefficients of x^2 and y^2 are both exactly 1 and there is no cross term xy, this rigorously proves it is the equation of a **circle** in the x-y plane.

7. **Finding the Center and Radius (Completing the Square):**
To find the explicit geometric parameters, complete the square for the x terms. Let $M = a\left(\frac{1+K^2}{1-K^2}\right)$. The equation is $x^2 + 2Mx + y^2 = -a^2$.

$$x^2 + 2Mx + M^2 + y^2 = M^2 - a^2 \tag{126}$$

$$(x+M)^2 + y^2 = M^2 - a^2 \tag{127}$$

This is the standard circle equation $(x-h)^2 + (y-k)^2 = R_{\text{circle}}^2$. The center coordinates (h, k) are:

$$h = -M = -a\left(\frac{1+K^2}{1-K^2}\right) = a\left(\frac{K^2+1}{K^2-1}\right), \quad k = 0 \tag{128}$$

The radius R_{circle} is:

$$R_{\text{circle}}^2 = \left[a\left(\frac{1+K^2}{1-K^2}\right)\right]^2 - a^2 = a^2\left[\frac{(1+K^2)^2 - (1-K^2)^2}{(1-K^2)^2}\right] \tag{129}$$

$$R_{\text{circle}}^2 = a^2\left[\frac{(1+2K^2+K^4) - (1-2K^2+K^4)}{(1-K^2)^2}\right] = a^2\left[\frac{4K^2}{(1-K^2)^2}\right] \tag{130}$$

$$R_{\text{circle}} = \frac{2aK}{K^2-1} \tag{131}$$

Final Answer

Final Conclusion of Proof:
The algebraic derivation strictly yields $(x-h)^2 + y^2 = R^2$, which is the locus of a circle in a 2D cross-section. Because the physical line charges are infinitely long, extending uniformly along the $\pm z$ axis (orthogonal to the page), these 2D circular cross-sections physically sweep out three-dimensional **cylinders** parallel to the z-axis. Thus, it is definitively proven that the equipotential surfaces are a family of non-concentric cylinders.

5.2 (b) Validity of a Specific Vector Field (4 Marks)

Problem Statement:
Determine whether $\mathbf{E} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ is a valid form of electric field or not.

Detailed Verification:
For any arbitrary mathematical vector field to physically represent a valid *static* electrostatic field, it must obey Maxwell's equations for electrostatics. The most stringent constraint is

Faraday's Law of Induction for static fields ($\partial \mathbf{B}/\partial t = 0$), which dictates that an electrostatic field must be conservative (irrotational).

Mathematically, a vector field is conservative if and only if its curl is exactly zero everywhere in space:

$$\nabla \times \mathbf{E} = \mathbf{0} \tag{132}$$

If this holds true, it guarantees that the work done moving a charge around any closed loop is zero ($\oint \mathbf{E} \cdot d\mathbf{l} = 0$), conserving energy.

Let us compute the curl of the given field $\mathbf{E} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$.

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} \tag{133}$$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} \tag{134}$$

Expanding the determinant by the top row (unit vectors):

$$\nabla \times \mathbf{E} = \hat{\mathbf{i}} \left(\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(y) \right) - \hat{\mathbf{j}} \left(\frac{\partial}{\partial x}(z) - \frac{\partial}{\partial z}(x) \right) + \hat{\mathbf{k}} \left(\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) \right) \tag{135}$$

Since x, y, and z are mutually independent spatial variables, the partial derivative of any one variable with respect to a different variable is intrinsically zero.

$$\nabla \times \mathbf{E} = \hat{\mathbf{i}}(0-0) - \hat{\mathbf{j}}(0-0) + \hat{\mathbf{k}}(0-0) = 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + 0\hat{\mathbf{k}} = \mathbf{0} \tag{136}$$

Final Answer

Conclusion:
Because the curl of the vector field $\nabla \times \mathbf{E}$ evaluates identically to zero, the field is conservative and irrotational. Therefore, $\mathbf{E} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ **is a perfectly valid form** of an electrostatic field.
Bonus Insight: Taking the divergence ($\boldsymbol{\nabla} \cdot E = 1 + 1 + 1 = 3$) reveals that this specific field corresponds physically to a region containing a completely uniform volume charge density $\rho_v = 3\epsilon_0$.

5.3 (c) Economical Core Radius in a Single Core Cable (4 Marks)

Problem Statement:
In the case of a single core single dielectric cable... determine the value of electric field intensity on the inner conductor of radius r for most economical use...

Detailed Optimization Derivation:

In a high-voltage coaxial cable, the geometry consists of a central cylindrical copper/aluminum core of radius r, surrounded by solid dielectric insulation, which is then encased by an outer grounded metallic sheath of inner radius R.

The electric field intensity (dielectric stress E) at any radial distance x (where $r \leq x \leq R$) within the dielectric is derived from Gauss's law and potential difference, giving:

$$E_x = \frac{V}{x \ln(R/r)} \tag{137}$$

where V is the operating voltage applied to the core.

Because the field is inversely proportional to x, the maximum dielectric stress (E_{max}) always occurs exactly at the surface of the inner conductor where x is at its minimum value ($x = r$):

$$E_{\text{max}} = \frac{V}{r\ln(R/r)} \tag{138}$$

Defining the "Most Economical Use":
The overarching goal of economical cable design is to minimize the total cross-sectional area (and thus material cost) while preventing insulation breakdown. For a fixed operating voltage V and a fixed outer sheath size R (dictated by trench/duct sizing), we must choose the core radius r that inherently *minimizes* the peak dielectric stress E_{max}. Minimizing E_{max} means we can use thinner or cheaper insulation materials.

To find the minimum value of E_{max} with respect to the variable r, we apply calculus optimization: we take the first derivative of E_{max} with respect to r and set it to zero.

$$\frac{dE_{\text{max}}}{dr} = \frac{d}{dr}\left[\frac{V}{r\ln(R/r)}\right] = 0 \tag{139}$$

Since the numerator V is a constant, to minimize the fraction we must **maximize** the denominator function $f(r) = r\ln(R/r)$.

$$\frac{d}{dr}[r\ln(R/r)] = 0 \tag{140}$$

Apply the product rule ($d(uv) = u'v + uv'$), where $u = r$ and $v = \ln(R/r)$:

$$\frac{d}{dr}[r\ln(R/r)] = (1)\cdot\ln\left(\frac{R}{r}\right) + r\cdot\frac{d}{dr}\left[\ln(R/r)\right] \tag{141}$$

$$= \ln\left(\frac{R}{r}\right) + r\cdot\left(\frac{1}{R/r}\right)\cdot\frac{d}{dr}\left(\frac{R}{r}\right) \tag{142}$$

$$= \ln\left(\frac{R}{r}\right) + r\cdot\left(\frac{r}{R}\right)\cdot\left(-\frac{R}{r^2}\right) \tag{143}$$

$$= \ln\left(\frac{R}{r}\right) - \left(\frac{r^2R}{Rr^2}\right) \tag{144}$$

$$= \ln\left(\frac{R}{r}\right) - 1 = 0 \tag{145}$$

Solving for the optimal ratio:

$$\ln\left(\frac{R}{r}\right) = 1 \tag{146}$$

$$e^{\ln(R/r)} = e^1 \tag{147}$$

$$\frac{R}{r} = e \approx 2.718 \tag{148}$$

This ratio ($R/r = 2.718$) represents the most economical geometric design for a single-core cable.

Finding the Field Intensity Value:
The question explicitly asks to determine the *value* of the electric field intensity on the inner conductor under this optimized condition. We substitute the optimal condition $\ln(R/r) = 1$ back into the original maximum stress equation:

$$E_{\text{max}} = \frac{V}{r\ln(R/r)} \tag{149}$$

$$E_{\text{max}} = \frac{V}{r(1)} \tag{150}$$

$$E_{\text{max}} = \frac{V}{r} \tag{151}$$

Alternatively, substituting $r = R/e$:

$$E_{\max} = \frac{V}{R/e} = \frac{eV}{R} \tag{152}$$

Final Answer

For the most economical use of the cable, the electric field intensity on the inner conductor is strictly minimized and evaluates to:
$E_{\max} = \mathbf{\frac{V}{r}}$ (which is also equivalent to $\mathbf{\frac{eV}{R}}$).

PART II: Magnetostatics & Time-Varying Fields

6 Question 1 - Divergence, Curl, and Magnetic Fields

6.1 (a) Fundamental Definition of Curl and Physical Significance

Definition:
The curl of a vector field $\mathbf{A}$, denoted as $\nabla \times \mathbf{A}$, is a vector whose magnitude is the maximum macroscopic circulation (line integral) of $\mathbf{A}$ per unit area as the area tends to zero, and whose direction is the normal direction of the area when the area is oriented to make the circulation maximum.

$$\nabla \times \mathbf{A} = \left(\lim_{\Delta S \to 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \hat{\mathbf{n}} \tag{153}$$

Physical Significance:
Curl physically represents the "rotation" or "vorticity" of a field at a localized point. For example, if a tiny paddlewheel placed in a flowing fluid spins, the fluid field has non-zero curl at that point. In electromagnetics, a non-zero curl of magnetic field ($\nabla \times \mathbf{H} = \mathbf{J}$) means the magnetic field lines are actively forming closed loops around a current source $\mathbf{J}$.

6.2 (b) Derivation of $\boldsymbol{\nabla} \cdot B = 0$

The fundamental postulate of magnetostatics states that there are no isolated magnetic monopoles (no single North or South poles exist). Magnetic field lines strictly form continuous, unbroken loops. Applying Gauss's Law to a magnetic field over any closed surface S:

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \tag{154}$$

Applying the Divergence Theorem to convert the surface integral to a volume integral:

$$\int_v (\boldsymbol{\nabla} \cdot B)\, dv = 0 \tag{155}$$

Because this must hold true for absolutely any arbitrary volume element v, the integrand itself must identically vanish everywhere:

$$\boldsymbol{\nabla} \cdot B = 0 \tag{156}$$

This differential Maxwell equation proves magnetic fields are solenoidal (divergence-less).

7 Question 2 - Vector Theorems and Maxwell's Equations

7.1 (a) Stoke's Theorem

Statement:
Stokes' Theorem states that the open surface integral of the curl of a vector field $\mathbf{A}$ over any

surface S is exactly equal to the closed line integral (circulation) of the vector field $\mathbf{A}$ along the bounding contour L that forms the perimeter of surface S.

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_L \mathbf{A} \cdot d\mathbf{l} \tag{157}$$

7.2 (b) Establish $\nabla \times \mathbf{H} = \mathbf{J}$ (Ampere's Law)

Start with Ampere's Circuital Law in integral form, which states the line integral of magnetic field intensity $\mathbf{H}$ around a closed path L equals the total direct current I_{enc} enclosed by the path:

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \tag{158}$$

The enclosed current can be written as the surface integral of the current density vector $\mathbf{J}$ over the area S bounded by L:

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{s} \tag{159}$$

Substitute this into Ampere's Law:

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{s} \tag{160}$$

Now, apply Stokes' Theorem to the left side to convert the line integral of $\mathbf{H}$ to a surface integral of its curl:

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s} \tag{161}$$

Since this holds for any arbitrary open surface S, the integrands must be identical. Thus, we arrive at the differential form of Ampere's Law:

$$\nabla \times \mathbf{H} = \mathbf{J} \tag{162}$$

7.3 (c) Establish $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ (Faraday's Law)

Faraday's Law of Induction in integral form states that the induced electromotive force (EMF) in a closed loop L is equal to the negative time rate of change of magnetic flux Φ_B passing through the surface S bounded by L:

$$\text{EMF} = -\frac{d\Phi_B}{dt} \tag{163}$$

By definition, EMF is the work done per unit charge, which is the closed line integral of the electric field: $\text{EMF} = \oint_L \mathbf{E} \cdot d\mathbf{l}$. Magnetic flux is the surface integral of magnetic flux density: $\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{s}$. Substituting these yields:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \tag{164}$$

Assuming a stationary loop, the time derivative can move inside the integral as a partial derivative:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \int_S \left(-\frac{\partial \mathbf{B}}{\partial t}\right) \cdot d\mathbf{s} \tag{165}$$

Apply Stokes' Theorem to the left side:

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = \int_S \left(-\frac{\partial \mathbf{B}}{\partial t}\right) \cdot d\mathbf{s} \tag{166}$$

Equating the integrands for any arbitrary surface yields the point form of Faraday's Law:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{167}$$

8 Question 4 - Poynting Theorem and Biot-Savart Applications

8.1 (a) Establish "Poynting Theorem"

Poynting Theorem is the statement of energy conservation for electromagnetic fields. We begin with the two curl Maxwell equations for time-varying fields:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad \text{(Faraday)} \tag{168}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \quad \text{(Ampere-Maxwell)} \tag{169}$$

Take the dot product of $\mathbf{E}$ with Ampere's Law:

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{E} \cdot (\sigma \mathbf{E}) + \mathbf{E} \cdot \left(\epsilon \frac{\partial \mathbf{E}}{\partial t} \right) \tag{170}$$

Use the vector identity: $\boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$. Let $\mathbf{A} = \mathbf{E}$ and $\mathbf{B} = \mathbf{H}$:

$$\boldsymbol{\nabla} \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}) \tag{171}$$

Rearranging for $\mathbf{E} \cdot (\nabla \times \mathbf{H})$:

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \boldsymbol{\nabla} \cdot (\mathbf{E} \times \mathbf{H}) \tag{172}$$

Substitute Faraday's law ($\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$) into this:

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{H} \cdot \left(-\mu \frac{\partial \mathbf{H}}{\partial t} \right) - \boldsymbol{\nabla} \cdot (\mathbf{E} \times \mathbf{H}) \tag{173}$$

Now equate the two expressions we have for $\mathbf{E} \cdot (\nabla \times \mathbf{H})$:

$$-\mu \mathbf{H} \cdot \frac{\partial \mathbf{H}}{\partial t} - \boldsymbol{\nabla} \cdot (\mathbf{E} \times \mathbf{H}) = \sigma |\mathbf{E}|^2 + \epsilon \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \tag{174}$$

Use the derivative property $\mathbf{A} \cdot \frac{\partial \mathbf{A}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} |\mathbf{A}|^2$:

$$-\frac{1}{2} \mu \frac{\partial |\mathbf{H}|^2}{\partial t} - \boldsymbol{\nabla} \cdot (\mathbf{E} \times \mathbf{H}) = \sigma |\mathbf{E}|^2 + \frac{1}{2} \epsilon \frac{\partial |\mathbf{E}|^2}{\partial t} \tag{175}$$

Rearranging and grouping the time derivatives:

$$-\boldsymbol{\nabla} \cdot (\mathbf{E} \times \mathbf{H}) = \sigma |\mathbf{E}|^2 + \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\mathbf{E}|^2 + \frac{1}{2} \mu |\mathbf{H}|^2 \right) \tag{176}$$

Integrate this entire equation over a volume v bounded by a closed surface S, and apply the divergence theorem to the left side:

$$-\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{s} = \int_v \sigma |\mathbf{E}|^2 \, dv + \frac{\partial}{\partial t} \int_v \left(\frac{1}{2} \epsilon |\mathbf{E}|^2 + \frac{1}{2} \mu |\mathbf{H}|^2 \right) dv \tag{177}$$

Let the Poynting Vector be defined as $\mathbf{P} = \mathbf{E} \times \mathbf{H}$ (in W/m^2). This final integral equation is the **Poynting Theorem**, proving that the net electromagnetic power flowing *into* a volume (left side) is equal to the power dissipated as ohmic heat (first term right) plus the rate of increase of stored electric and magnetic energy (second term right).

APPENDICES: Formula References

A Maxwell's Equations Summary

Name	Differential Form	Integral Form
Gauss's Law for Electricity	$\boldsymbol{\nabla} \cdot D = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_v \rho_v \, dv$
Gauss's Law for Magnetism	$\boldsymbol{\nabla} \cdot B = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$
Faraday's Law of Induction	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$
Ampere-Maxwell Law	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}\right) \cdot d\mathbf{s}$

Table 1: Summary of Maxwell's Equations in both differential and integral forms.

B Coordinate Systems Transformations

1. Cartesian to Cylindrical:

$$r = \sqrt{x^2 + y^2} \tag{178}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \tag{179}$$

$$z = z \tag{180}$$

2. Cylindrical to Cartesian:

$$x = r\cos(\phi) \tag{181}$$

$$y = r\sin(\phi) \tag{182}$$

$$z = z \tag{183}$$

3. Cartesian to Spherical:

$$R = \sqrt{x^2 + y^2 + z^2} \tag{184}$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \tag{185}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \tag{186}$$